

The Ghosh Solution of the Vacuum Ernst Equation

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

This paper develops a non-trivial exact solution of the stationary, axisymmetric vacuum Ernst equation by combining the structural relations introduced in Ghosh's construction of a universal constant with the corresponding energy relation for a uniformly dense spherical black hole. The Ghosh framework introduces the quantity

$$\Upsilon = \frac{3c}{4\pi G}, \quad (1)$$

and, from the relations among density, radius, gravitational acceleration, and formation time, obtains

$$\rho RT = \Upsilon. \quad (2)$$

The associated energy relation,

$$E = \frac{64}{27} G^2 \pi^3 \Upsilon^2 \rho R^3, \quad (3)$$

provides the corresponding mass and gravitational length scales.

We combine these structural relations with the vacuum Ernst equation and impose the symmetry information inherent in the uniformly dense spherical construction. The resulting real Ernst potential reduces the nonlinear Ernst equation to a harmonic boundary-value problem in Weyl coordinates. Solving this problem for an asymptotically flat compact spherical source yields an explicit non-constant Ernst potential,

$$\mathcal{E}_G = \frac{R_+ + R_- - 2\mu_G}{R_+ + R_- + 2\mu_G}, \quad (4)$$

where

$$R_{\pm} = \sqrt{\rho_W^2 + (z \pm \mu_G)^2}, \quad (5)$$

and the characteristic gravitational scale is determined entirely by the Ghosh variables,

$$\mu_G = \frac{3\rho R^3}{4\pi G \Upsilon^2} = \frac{3R}{4\pi G \rho T^2}. \quad (6)$$

The construction demonstrates how the physical scaling relations of the Ghosh framework can serve as boundary and normalization data for an exact solution of the vacuum Ernst equation. The resulting geometry belongs to the static Schwarzschild sector; the significance of the construction therefore lies not in asserting a new inequivalent vacuum spacetime, but in deriving an exact Ernst solution whose characteristic gravitational scale is expressed directly through the Ghosh variables.

The paper ends with “The End”

1. The Question

The problem is to determine what structural information contained in the two Ghosh papers can be combined legitimately with the stationary, axisymmetric vacuum Ernst equation in order to construct a non-trivial solution, without simply declaring the answer to be Schwarzschild.

The essential point is that the papers contain more than the isolated definition of the quantity

$$\Upsilon = \frac{3c}{4\pi G}. \quad (7)$$

They specify a physical structure:

$$\boxed{\text{uniform density} + \text{spherical geometry} + \text{radius} + \text{mass} + \text{surface gravity} + \text{formation time.}} \quad (8)$$

The first paper establishes

$$\rho = \frac{M}{\frac{4}{3}\pi R^3}, \quad (9)$$

$$g = \frac{GM}{R^2}, \quad (10)$$

and

$$c = gT, \quad (11)$$

from which

$$\boxed{\rho RT = \Upsilon} \quad (12)$$

follows.

The second paper supplements this with

$$E = Mc^2 \quad (13)$$

and obtains

$$\boxed{E = \frac{64}{27}G^2\pi^3\Upsilon^2\rho R^3.} \quad (14)$$

The task is therefore to determine what these relations imply when the exterior region is required to satisfy the vacuum Ernst equation.

2. The Structural Content of the Ghosh Relations

From

$$\rho = \frac{3M}{4\pi R^3}, \quad (15)$$

we recover

$$\boxed{M = \frac{4\pi}{3}\rho R^3.} \quad (16)$$

The relation

$$\rho RT = \Upsilon \quad (17)$$

gives

$$\boxed{T = \frac{\Upsilon}{\rho R}.} \quad (18)$$

The second paper gives

$$E = \frac{64}{27}G^2\pi^3\Upsilon^2\rho R^3. \quad (19)$$

Since

$$E = Mc^2, \quad (20)$$

the mass scale is therefore not independent of the Ghosh variables.

Using

$$c = \frac{4\pi G}{3}\Upsilon, \quad (21)$$

we obtain

$$M = \frac{E}{c^2} \quad (22)$$

$$= \frac{\frac{64}{27} G^2 \pi^3 \Upsilon^2 \rho R^3}{\frac{16}{9} G^2 \pi^2 \Upsilon^2} \quad (23)$$

$$= \frac{4\pi}{3} \rho R^3. \quad (24)$$

Thus the two papers are mutually consistent.

3. What Can Enter the Ernst Equation?

The stationary, axisymmetric vacuum Ernst equation is

$$(\text{Re } \mathcal{E}) \nabla^2 \mathcal{E} = \nabla \mathcal{E} \cdot \nabla \mathcal{E}, \quad (25)$$

where the differential operator acts on the Weyl coordinates

$$(\rho_W, z). \quad (26)$$

The structural information supplied by the Ghosh papers has an important consequence.

The source is described as a uniformly dense sphere.

Therefore there is no preferred direction in the source interior.

Consequently, the exterior gravitational field has no angular-momentum parameter and no independently specified higher multipole moments in the supplied data.

Thus the Ghosh data determine a mass scale but do not provide a twist source.

This means that the natural Ernst potential is real:

$$\boxed{\text{Im } \mathcal{E} = 0.} \quad (27)$$

Equation (25) consequently becomes

$$\mathcal{E} \nabla^2 \mathcal{E} = \nabla \mathcal{E} \cdot \nabla \mathcal{E}. \quad (28)$$

Writing

$$\mathcal{E} = e^{2U}, \quad (29)$$

we obtain

$$\nabla \mathcal{E} = 2e^{2U} \nabla U, \quad (30)$$

$$\nabla^2 \mathcal{E} = 2e^{2U} \nabla^2 U + 4e^{2U} \nabla U \cdot \nabla U. \quad (31)$$

Substitution into the Ernst equation gives

$$e^{2U} [2e^{2U} \nabla^2 U + 4e^{2U} (\nabla U)^2] = 4e^{4U} (\nabla U)^2. \quad (32)$$

The gradient-square terms cancel, leaving

$$\boxed{\nabla^2 U = 0.} \quad (33)$$

Thus the Ghosh structural assumptions reduce the nonlinear Ernst problem to a harmonic boundary-value problem.

4. The Ghosh Boundary Structure

The remaining information is that the source possesses a characteristic radius and mass.

In Weyl coordinates, the natural static axisymmetric vacuum solution corresponding to a compact spherical source is generated by a finite interval on the symmetry axis.

Let the half-length of this interval be

$$\mu. \quad (34)$$

The associated distances are

$$R_{\pm} = \sqrt{\rho_W^2 + (z \pm \mu)^2}. \quad (35)$$

The harmonic potential generated by the finite interval is

$$U = \frac{1}{2} \ln \left(\frac{R_+ + R_- - 2\mu}{R_+ + R_- + 2\mu} \right). \quad (36)$$

Because U is harmonic away from the source interval,

$$\nabla^2 U = 0, \quad (37)$$

and therefore

$$\mathcal{E} = e^{2U} \quad (38)$$

satisfies the Ernst equation.

Hence

$$\mathcal{E} = \frac{R_+ + R_- - 2\mu}{R_+ + R_- + 2\mu}. \quad (39)$$

At this stage we have not named the solution.

We have derived its functional form from:

staticity,
 axisymmetry,
 vacuum exterior,
 asymptotic flatness,
 compact spherical source.

(40)

5. Determining the Scale from the Ghosh Papers

The remaining question is how the Ghosh quantities determine μ .

The relativistic length scale associated with the mass is

$$\mu = \frac{GM}{c^2}. \quad (41)$$

Using

$$M = \frac{4\pi}{3} \rho R^3, \quad (42)$$

we obtain

$$\mu = \frac{4\pi G \rho R^3}{3c^2}. \quad (43)$$

Using

$$c = \frac{4\pi G}{3} \Upsilon, \quad (44)$$

gives

$$\mu = \frac{4\pi G \rho R^3}{3 \left(\frac{16\pi^2 G^2}{9} \right) \Upsilon^2} \quad (45)$$

$$= \boxed{\frac{3\rho R^3}{4\pi G \Upsilon^2}}. \quad (46)$$

Using instead

$$\Upsilon = \rho R T, \quad (47)$$

we obtain

$$\mu = \frac{3\rho R^3}{4\pi G \rho^2 R^2 T^2} \quad (48)$$

$$= \boxed{\frac{3R}{4\pi G \rho T^2}}. \quad (49)$$

Thus the Ghosh relations determine the sole dimensional scale appearing in the vacuum Ernst solution.

6. The Derived Ghosh Ernst Potential

Substitution of (46) into (39) gives

$$\mathcal{E}_G = \frac{R_+ + R_- - \frac{3\rho R^3}{2\pi G \Upsilon^2}}{R_+ + R_- + \frac{3\rho R^3}{2\pi G \Upsilon^2}}, \quad (50)$$

where

$$R_{\pm} = \sqrt{\rho_W^2 + \left(z \pm \frac{3\rho R^3}{4\pi G \Upsilon^2} \right)^2}. \quad (51)$$

This is a non-constant exact solution of the Ernst equation.

Importantly, it has not been introduced by simply declaring it to be Schwarzschild.

It has been obtained by solving the Ernst boundary-value problem after importing the structural scale supplied by the Ghosh relations.

7. Equivalent Ghosh-Time Representation

Using

$$\Upsilon = \rho R T, \quad (52)$$

the characteristic length becomes

$$\mu = \frac{3R}{4\pi G \rho T^2}. \quad (53)$$

Therefore the solution can be written as

$$\mathcal{E}_G = \frac{R_+ + R_- - \frac{3R}{2\pi G \rho T^2}}{R_+ + R_- + \frac{3R}{2\pi G \rho T^2}}, \quad (54)$$

with

$$R_{\pm} = \sqrt{\rho_W^2 + \left(z \pm \frac{3R}{4\pi G \rho T^2}\right)^2}. \quad (55)$$

Thus the same exact solution may be parameterized by either

$$(\rho, R, \Upsilon) \quad (56)$$

or

$$(\rho, R, T). \quad (57)$$

8. Why This Is Not Merely a Declaration of Schwarzschild

The logical order is important.

We have not started with the proposition

$$\text{“the solution is Schwarzschild.”} \quad (58)$$

Instead, we began with

$$\text{Ghosh structural assumptions} \longrightarrow \text{Im } \mathcal{E} = 0, \quad (59)$$

then obtained

$$\text{Re } \mathcal{E} = e^{2U}, \quad (60)$$

then derived

$$\nabla^2 U = 0, \quad (61)$$

then imposed the compact spherical boundary structure and asymptotic flatness, obtaining

$$U = \frac{1}{2} \ln \left(\frac{R_+ + R_- - 2\mu}{R_+ + R_- + 2\mu} \right), \quad (62)$$

and only then determined

$$\mu = \frac{3\rho R^3}{4\pi G \Upsilon^2}. \quad (63)$$

The known spacetime identification is therefore a *consequence* rather than an input.

9. What the Ghosh Papers Do and Do Not Add

There is an important distinction between a new solution and a new parameterization.

The Ghosh papers provide the relation

$$\Upsilon = \frac{3c}{4\pi G}, \quad (64)$$

and therefore provide a particular parameterization of the gravitational scale.

They do not provide:

a non-zero twist potential, (65)

an angular momentum parameter, (66)

a quadrupole or higher multipole moment, (67)

a second independent gravitational length, (68)

or a modified vacuum field equation. (69)

Consequently, the Ghosh information alone does not generate a new inequivalent stationary vacuum geometry.

Instead, it generates a particular parameterization of the unique static, spherically symmetric vacuum sector selected by those assumptions.

10. The Deeper Structural Result

The genuinely interesting result is therefore not

“Ghosh discovered Schwarzschild.” (70)

Nor is it

“there is no connection between Ghosh’s equations and Ernst’s equation.” (71)

The correct structural statement is

Ghosh’s relations determine the scale parameter of an Ernst boundary-value problem.

 (72)

More explicitly,

$$\boxed{\Upsilon \longrightarrow c \longrightarrow M \longrightarrow \mu \longrightarrow \mathcal{E}_G.} \quad (73)$$

The first paper supplies the relation between the density, radius and formation time,

$$\rho RT = \Upsilon, \quad (74)$$

while the second supplies the corresponding energy scale.

The Ernst equation supplies the nonlinear gravitational field structure.

Thus the three ingredients play different mathematical roles:

Ingredient	Mathematical role	Physical role
Υ	Parameterization	Universal scale
ρ, R, T	Boundary data	Source structure
$E = Mc^2$	Mass normalization	Energy scale
Ernst equation	Field equation	Vacuum geometry
Spherical symmetry	Symmetry reduction	No rotation/twist
Asymptotic flatness	Boundary condition	Isolated system

11. Final Answer

The structural information that can legitimately be imported from the two Ghosh papers is therefore

$$\rho = \frac{3M}{4\pi R^3}, \quad g = \frac{GM}{R^2}, \quad c = gT, \quad \Upsilon = \frac{3c}{4\pi G}, \quad (75)$$

together with

$$E = Mc^2 = \frac{64}{27}G^2\pi^3\Upsilon^2\rho R^3. \quad (76)$$

Combined with the static, axisymmetric vacuum Ernst equation, these imply a real Ernst potential generated by a harmonic Weyl potential.

The resulting exact non-trivial solution is

$$\mathcal{E}_G = \frac{R_+ + R_- - 2\mu_G}{R_+ + R_- + 2\mu_G}, \quad (77)$$

where

$$\mu_G = \frac{3\rho R^3}{4\pi G\Upsilon^2} = \frac{3R}{4\pi G\rho T^2}, \quad (78)$$

and

$$R_{\pm} = \sqrt{\rho_W^2 + (z \pm \mu_G)^2}. \quad (79)$$

This solution is non-constant and satisfies the Ernst equation by construction.

Only after deriving it may we identify the resulting vacuum geometry with the Schwarzschild sector.

Thus the mathematically precise conclusion is

The Ghosh papers do provide legitimate structural input to the Ernst problem;
 that input fixes the mass/length scale of the solution;
 the Ernst equation determines the vacuum field structure;
 but the supplied Ghosh papers do not introduce a new independent
 degree of freedom capable of producing a geometry inequivalent to Schwarzschild.

(80)

In other words, the correct achievement is not the invention of an unsupported “Ghosh space-time.” It is the derivation of a *Ghosh-parameterized exact Ernst solution* from the structural assumptions of the two papers.

The End